EASY ALGEBRA

Linear equations in two variables

01

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 ALGEBRA

A teacher is creating an assignment worth 70 points. The assignment will consist of questions worth 1 point and questions worth 3 points. Which equation represents this situation, where x represents the number of 1-point questions and y represents the number of 3-point questions?

A. $4xy = 70$

B. $4x + y = 70$

C. $3x + y = 70$

D. $x + 3y = 70$

Q2 ALGEBRA

A line in the xy-plane has a slope of $\frac{1}{9}$ and passes through the point $0, 14$. Which equation represents this line?

A. $y = -\frac{1}{9}x - 14$

B. $y = -\frac{1}{9}x + 14$

C. $y = \frac{1}{9}x - 14$

D. $y = \frac{1}{9}x + 14$

Q3

GRID-IN ALGEBRA

The equation $46 = 2a + 2b$ gives the relationship between the side lengths a and b of a certain parallelogram. If $a = 9$, what is the value of b?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4 ALGEBRA

Jay walks at a speed of 3 miles per hour and runs at a speed of 5 miles per hour. He walks for w hours and runs for r hours for a combined total of 14 miles. Which equation represents this situation?

A. $3w + 5r = 14$

B. $\frac{1}{3}w + \frac{1}{5}r = 14$

C. $\frac{1}{3}w + \frac{1}{5}r = 112$

D. $3w + 5r = 112$

Q5 ALGEBRA

$$y = 70x + 8$$

Which table gives three values of x and their corresponding values of y for the given equation?

A.

x	y
0	8
2	148
4	288

B.

x	y
0	70
2	78
4	86

C.

x	y
0	70
2	140
4	280

D.

x	y
0	8
2	132
4	272

Q6 ALGEBRA

A line in the xy-plane has a slope of $-\frac{1}{2}$ and passes through the point $0, 3$. Which equation represents this line?

A. $y = -\frac{1}{2}x - 3$

B. $y = -\frac{1}{2}x + 3$

C. $y = \frac{1}{2}x - 3$

D. $y = \frac{1}{2}x + 3$

Q7

ALGEBRA

What is the equation of the line that passes through the point $0, 5$ and is parallel to the graph of $y = 7x + 4$ in the xy-plane?

A. $y = 5x$

B. $y = 7x + 5$

C. $y = 7x$

D. $y = 5x + 7$

Q8 ALGEBRA

In the xy-plane, a line has a slope of 6 and passes through the point $(0,8)$. Which of the following is an equation of this line?

A. $y=6x+8$

B. $y=6x+48$

C. $y=8x+6$

D. $y=8x+48$

Q9 GRID-IN ALGEBRA

Line k is defined by $y = \frac{1}{4}x + 1$. Line j is parallel to line k in the xy-plane. What is the slope of j?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q10 ALGEBRA

Naomi bought both rabbit snails and nerite snails for a total of $\$52$. Each rabbit snail costs $\$8$ and each nerite snail costs $\$6$. If Naomi bought 2 nerite snails, how many rabbit snails did she buy?

A. 5

B. 12

C. 14

D. 50